Lid-Driven Cavity Flow: A Finite-Volume Implementation

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1 Problem Description

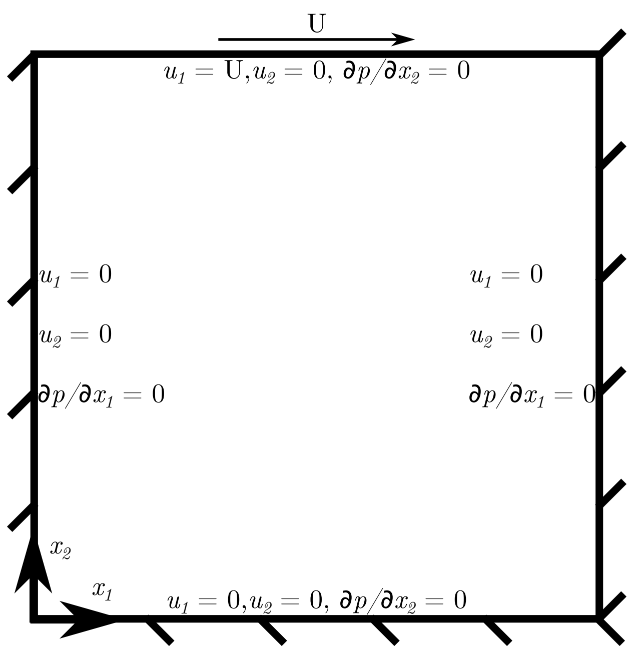

Figure 1: Lid-driven cavity flow problem illustration.

2 Governing Equations

2.1 Differential Equations

The form of the incompressible Navier-Stokes equations, and their subsequent non-dimensionalization are taken from Ferziger and Perić [1]. Dimensional quantities are shown with a "*" superscript, and non-dimensional quantities are shown without modification. Conservation of mass, with dimensional units is shown in Eq. 1. Similarly, the dimensional form of the momentum equation is given in Eq. 2.

$$\frac{\partial u_i^*}{\partial x_i^*} = 0 \tag{1}$$

$$\frac{\partial u_i^*}{\partial t^*} + \frac{\partial (u_j^* u_i^*)}{\partial x_j^*} = \nu^* \frac{\partial^2 u_i^*}{\partial x_j^2} - \frac{1}{\rho^*}\frac{\partial p^*}{\partial x_i^*} \tag{2}$$

The dimensional quantities of Eq. 1, 2, are nondimensionalized according to Eqs. 3 - 6.

$$t = \frac{t^*}{t_0^*} \tag{3}$$

$$x_i = \frac{x_i^*}{L_0^*} \tag{4}$$

$$u_i^* = \frac{u_i^*}{v_0^*} \tag{5}$$

$$p = \frac{p^*}{\rho^* v_0^{*2}} \tag{6}$$

Accordingly, the nondimensional form of the continuity and momentum equation are shown in Eq. 7 and 8, respectively. In these equations, L_0^* is the reference length dimension, t_0^* is the reference time dimension, v_0^* is the refence velocity dimension, and ρ^* is the density of the fluid.

$$\frac{\partial u_i}{\partial x_i} = 0 \tag{7}$$

$$\boxed{\mathrm{St}\frac{\partial u_i}{\partial t} + \frac{\partial (u_j u_i)}{\partial x_j} = \frac{1}{\mathrm{Re}}\frac{\partial^2 u_i}{\partial x_j} - \frac{\partial p}{\partial x_i}} \tag{8}$$

As a result of the nondimensionalization, the dimensionless numbers, Re (Reynolds number) and St (Strouhal number) appear in Eq. 8. While Eqs. 7 and 8 mathematically express mass and momentum conservation, respectively, with regards to implementing these numerically, the continuity equation (Eq. 7) does not provide any additional information that is not contained in the momentum equation (Eq. 8). The continuity equation must be further manipulated into a form which can be implemented to enforce mass continuity. Computing the divergence of the momentum equation (Eq. 8) yields Eq. 9. Simplifying Eq. 9 produces equation Eq. 10. Applying the contuinuity equation to Eq. 10 yeilds, Eq. 11, the pressure Poisson equation. In solving the pressure Poisson equation, mass conservation is enforced. Note that there is not a fluid equation of state for closure in which *absolute* pressure appears as a term. In incompressible flow, only *gradients* of pressure appear in the momentum equation, thus by solving Eq. 11, a pressure field which satisfies continuity is determined.

$$\frac{\partial}{\partial x_i}\left(\mathrm{St}\frac{\partial u_i}{\partial t}\right) + \frac{\partial}{\partial x_i}\left(\frac{\partial (u_j u_i)}{\partial x_j}\right) = \frac{\partial}{\partial x_i}\left(\frac{1}{\mathrm{Re}}\frac{\partial^2 u_i}{\partial x_j}\right) - \frac{\partial}{\partial x_i}\left(\frac{\partial p}{\partial x_i}\right) \tag{9}$$

$$\mathrm{St}\frac{\partial}{\partial t}\left(\frac{\partial u_i}{\partial x_i}\right) + \frac{\partial}{\partial x_i}\left(\frac{\partial (u_j u_i)}{\partial x_j}\right) = \frac{1}{\mathrm{Re}}\frac{\partial}{\partial x_j}\left(\frac{\partial}{\partial x_j}\left(\frac{\partial u_i}{\partial x_i}\right)\right) - \frac{\partial}{\partial x_i}\left(\frac{\partial p}{\partial x_i}\right) \tag{10}$$

$$\boxed{\frac{\partial}{\partial x_i}\left(\frac{\partial (u_j u_i)}{\partial x_j}\right) = -\frac{\partial}{\partial x_i}\left(\frac{\partial p}{\partial x_i}\right)} \tag{11}$$

2.2 Integral Equations

The integral forms of the non-dimensional momentum equation (Eq. 8) is required to apply a finite volume method numerical solution. The integral form of the non-dimensional momentum equation is obtained by integrating the differential form of the equation over a volume domain, Ω, which yields Eq. 12. Gauss's Divergence Theorem is applied to Eq. 12 to transform the volume integrals over a volume domain Ω into surface integrals over a surface boundary σ. Furthermore, the constant non-dimensional terms are moved outside of the integrals, producing Eq. 13. This integral form of the nondimensional pressure equation, Eq. 13 is used for solving control volume problem, and thus is intuitive for applying to the uniform two-dimensional cells of this problem. The non-dimensional pressure Poisson equation, Eq. 11, is left in differential form.

$$\int_\Omega \mathrm{St}\frac{\partial u_i}{\partial t}d\Omega + \int_\Omega \frac{\partial (u_j u_i)}{\partial x_j}d\Omega = \int_\Omega \frac{1}{\mathrm{Re}}\frac{\partial^2 u_i}{\partial x_j^2}d\Omega - \int_\Omega \frac{\partial p}{\partial x_i}d\Omega \tag{12}$$

$$\boxed{\mathrm{St}\int_\Omega \frac{\partial u_i}{\partial t}d\Omega + \int_\sigma u_j u_i n_j d\sigma = \frac{1}{\mathrm{Re}}\int_\sigma \frac{\partial u_i}{\partial x_j}n_j d\sigma - \int_\sigma p n_j d\sigma} \tag{13}$$

3 Discretization

3.1 Domain and Grid

The domain of the problem depicted in Fig. 1 is described in dimensional and non-dimensional quantities in Eqs. 14, 15, respectively. A staggered grid is implemented, such that pressure nodes are placed in the center of the cells ($x_1 = x_{1,M}$, $x_2 = x_{2,M}$), u_1 nodes are placed at the mid-height of the west and east faces of the cells ($x_1 = x_1$, $x_2 = x_{2,M}$), and u_2 nodes are placed at the mid-width of the south and north faces of the cells ($x_1 = x_{1,M}$, $x_2 = x_2$), where the subscript "M" indicates "mid-point". The grid is uniform, as described in

Table 1: Coordinate locations for different node types.

Node type	x_1	x_2
u_1	$m\Delta x$	$n\frac{\Delta x}{2}$
u_2	$m\frac{\Delta x}{2}$	$n\Delta x$
p	$m\frac{\Delta x}{2}$	$n\frac{\Delta x}{2}$

Eq. 16. An index "m" is used for the x_1 coordinate, and an index "n" is used for the x_2 coordinate. The coordinates of the the u_1, u_2, and pressure (p) nodes are tabulated in Table 3.1, for clarity, and the grid is illustrated in Fig. 2. There are ghost-nodes outside of the boundary, in both the x_1 and x_2 directions to enforce boundary conditions.

$$x_1^* \in [0, L_0^*] x_2^* \in [0, L_0^*] \tag{14}$$

$$x_1 \in [0, 1] x_2 \in [0, 1] \tag{15}$$

$$\Delta x \equiv \Delta x_1 = \Delta x_2 \tag{16}$$

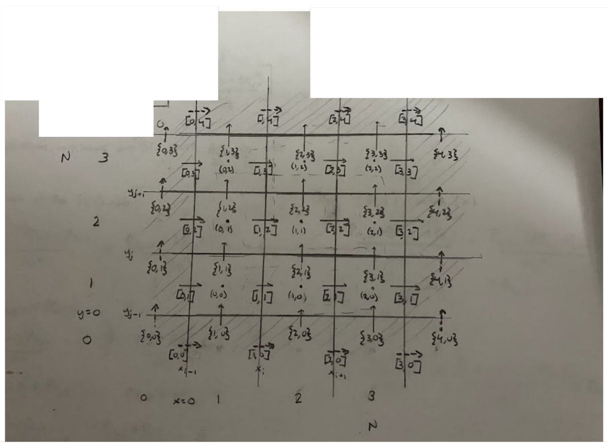

Figure 2: Illustration of staggered velocity-pressure grid.

3.2 Semi-Discrete Momentum Equation: Predictor and Corrector Step

The momentum equations, given in Eqs. 8, 13, cannot be solved in their given form as the pressure field that appears in these equations does not necessarily satisfy continuity. Accordingly, the momentum equation is split into a predictor step equation, Eq. 17, and a corrector step equation, Eq. 18, by discretizing the equations temporally. Note that "k" is the time-step index. An intermediate velocity field, u_i^*, is solved for in Eq. 17 which does not satisfy mass continuity. The pressure Poisson equation, 11, is then applied to this velocity field, u_i^*, as shown in Eq. 19, such that a pressure field, p, is solved for which does satisfy continuity. The corrector equation, 18, and a velocity field at time: $k+1$ is obtained which does satisfy mass continuity.

$$\mathrm{St}\frac{u_i^* - u_i^k}{\Delta t} = -\frac{\partial(u_j u_i)}{\partial x_j} + \frac{1}{\mathrm{Re}}\frac{\partial^2 u_i}{\partial x_j} \tag{17}$$

$$\mathrm{St}\frac{u_i^{k+1} - u_i^*}{\Delta t} = -\frac{\partial p}{\partial x_i} \tag{18}$$

$$\boxed{\frac{\partial}{\partial x_i}\left(\frac{\partial(u_j^* u_i^*)}{\partial x_j}\right) = -\frac{\partial}{\partial x_i}\left(\frac{\partial p}{\partial x_i}\right)} \tag{19}$$

Gauss's theorem is applied to re-cast the differential forms of the predictor step momentum equation, Eq. 17, and the corrector step momentum equation, 18 into integral equations for a finite volume approach, shown in Eqs. 20 and 21, respectively. In these equations, n, is the unit normal vector. The pressure Poisson equation is kept in differential form. Observe that on the right hand sides of Eqs. 20, 21 a temporal index, k, has been assigned which indicates that and *explicit* temporal discretization scheme will be implemented, meaning that the velocity field solution from the k time step will be used to solve for the velocity field at time $k+1$. The pressure field is not assigned a temporal index as it is merely a scalr field variable that is used to enforce mass continuity.

$$\boxed{\mathrm{St}\int_\Omega \frac{u_i^* - u_i^k}{\Delta t} d\Omega = -\int_\sigma u_j^k u_i^k n_j d\sigma + \frac{1}{\mathrm{Re}}\int_\sigma \frac{\partial u_i^k}{\partial x_j} n_j d\sigma} \tag{20}$$

$$\boxed{\mathrm{St}\int_\Omega \frac{u_i^{k+1} - u_i^*}{\Delta t} d\Omega = -\int_\sigma p n_i d\sigma} \tag{21}$$

3.3 Finite Volume Method and Spatial Discretization

The integral forms of the predictor momentum equation and corrector momentum equation are discretized in a finite volume method implementation in which the equations are organized into unsteady terms, convective flux, and diffusive flux terms, then numerically approximated as follows.

3.3.1 u_1 Momentum Equation

The unsteady term in the u_1 predictor momentum equation re-cast as shown in Eq. 22, where the velocity is approximated to be constant within the 2D cell, and thus the volume integral is simply the product of this velocity and the spatial dimensions of the cell.

$$\mathrm{St}\int_{\Omega}\frac{u_1^* - u_1^k}{\Delta t}d\Omega \approx \mathrm{St}\left[\frac{u_1^* - u_1^k}{\Delta t}\right]\Delta x_1 \Delta x_2 \equiv A_{u_1}[u_1^* - u_1^k] \tag{22}$$

The first term on the right hand side of the u_1 predictor equation represents the convective flux of velocity at the boundaries of the 2D cell, and is is evaluated on the σ_e (East), σ_w (West), σ_n (North), and σ_s (South) cell interfaces (walls), as shown in Eq. 23. Convective flux terms, such as $F_{c,e}^k$ are defined to be these area integrals, and are approximated by assuming a constant velocity on the cell face, as shown in Eqs. 24 - 27. In the 2D u_1 cells, the u_1 and u_2 velocites are not defined at the centers of the cell faces, and thus are approximated with 2nd-order accurate linear interpolation (averaging), shown in Eqs. 28 33.

$$\begin{aligned}\int_{\sigma} u_j^k u_i^k n_j d\sigma = \int_{\sigma} u_1^k u_1^k n_1 d\sigma + \int_{\sigma} u_2^k u_1^k n_2 d\sigma = \\ \int_{\sigma_e} u_1^k u_1^k d\sigma - \int_{\sigma_w} u_1^k u_1^k d\sigma + \int_{\sigma_n} u_2^k u_1^k d\sigma - \int_{\sigma_s} u_2^k u_1^k d\sigma\end{aligned} \tag{23}$$

$$\int_{\sigma} u_1^k u_1^k n_1 d\sigma \equiv F_{c,e}^k \approx u_{1,e}^k u_{1,e}^k \Delta x_2 \tag{24}$$

$$-\int_{\sigma_w} u_1^k u_1^k d\sigma \equiv F_{c,w}^k \approx -u_{1,w}^k u_{1,w}^k \Delta x_2 \tag{25}$$

$$\int_{\sigma_n} u_2^k u_1^k d\sigma \equiv F_{n,c}^k \approx u_{1,n}^k u_{2,n}^k \Delta x_1 \tag{26}$$

$$-\int_{\sigma_s} u_2^k u_1^k d\sigma \equiv F_{s,c}^k \approx -u_{1,s}^k u_{2,s}^k \Delta x_1 \tag{27}$$

$$u_{1,e}^k \approx \frac{u_{1,(m+1,n)}^k + u_{1,(m,n)}^k}{2} \tag{28}$$

$$u_{1,w}^k \approx \frac{u_{1,(m,n)}^k + u_{1,(m-1,n)}^k}{2} \tag{29}$$

$$u_{1,n}^k \approx \frac{u_{1,(m,n+1)}^k + u_{1,(m,n)}^k}{2} \tag{30}$$

$$u_{1,s}^k \approx \frac{u_{1,(m,n)}^k + u_{1,(m,n-1)}^k}{2} \tag{31}$$

$$u_{2,n}^k \approx \frac{u_{2,(m+1,n)}^k + u_{2,(m,n)}^k}{2} \tag{32}$$

$$u_{2,s}^k \approx \frac{u_{2,(m+1,n-1)}^k + u_{2,(m,n-1)}^k}{2} \tag{33}$$

The second term on the right hand side of the u_1 predictor equation represents the diffusive flux of velocity at the boundaries of the 2D cell, and is is evaluated on the σ_e (East), σ_w (West), σ_n (North), and σ_s (South) cell interfaces (walls), as shown in Eq. 34. Diffusive flux terms, such as $F_{d,e}^k$ are defined to be these area integrals, and are approximated by assuming a constant velocity on the cell face, as shown in Eqs. 35 - 38. The velocity derivatives at the cell face centers are approximated using 2nd-order accurate central differencing schemes, shown in Eqs. 39 - 42.

$$\begin{aligned}\frac{1}{\mathrm{Re}}\int_\sigma \frac{\partial u_1^k}{\partial x_j} n_j d\sigma = \frac{1}{\mathrm{Re}}\int_\sigma \frac{\partial u_1^k}{\partial x_1} n_1 d\sigma + \frac{1}{\mathrm{Re}}\int_\sigma \frac{\partial u_1^k}{\partial x_2} n_2 d\sigma = \\ \frac{1}{\mathrm{Re}}\int_{\sigma_e} \frac{\partial u_1^k}{\partial x_1} d\sigma - \frac{1}{\mathrm{Re}}\int_{\sigma_w} \frac{\partial u_1^k}{\partial x_1} d\sigma + \frac{1}{\mathrm{Re}}\int_{\sigma_n} \frac{\partial u_1^k}{\partial x_2} d\sigma - \frac{1}{\mathrm{Re}}\int_{\sigma_s} \frac{\partial u_1^k}{\partial x_2} d\sigma\end{aligned} \tag{34}$$

$$\frac{1}{\mathrm{Re}}\int_{\sigma_e} \frac{\partial u_1^k}{\partial x_1} d\sigma \equiv F_{d,e}^k \approx \frac{1}{\mathrm{Re}} \frac{\partial u_1}{\partial x_1}\bigg|_e^k \Delta x_2 \tag{35}$$

$$-\frac{1}{\mathrm{Re}}\int_{\sigma_e} \frac{\partial u_1^k}{\partial x_1} d\sigma \equiv F_{d,w}^k \approx -\frac{1}{\mathrm{Re}} \frac{\partial u_1}{\partial x_1}\bigg|_w^k \Delta x_2 \tag{36}$$

$$\frac{1}{\mathrm{Re}}\int_{\sigma_n} \frac{\partial u_1^k}{\partial x_2} d\sigma \equiv F_{d,n}^k \approx \frac{1}{\mathrm{Re}} \frac{\partial u_1}{\partial x_2}\bigg|_n^k \Delta x_1 \tag{37}$$

$$-\frac{1}{\mathrm{Re}}\int_{\sigma_s} \frac{\partial u_1^k}{\partial x_2} d\sigma \equiv F_{d,s}^k \approx -\frac{1}{\mathrm{Re}} \frac{\partial u_1}{\partial x_2}\bigg|_s^k \Delta x_1 \tag{38}$$

$$\frac{\partial u_1}{\partial x_1}\bigg|_e^k \approx \frac{u_{1,(m+1,n)}^k - u_{1,(m,n)}^k}{\Delta x_1} \tag{39}$$

$$\frac{\partial u_1}{\partial x_1}\bigg|_w^k \approx \frac{u_{1,(m,n)}^k - u_{1,(m-1,n)}^k}{\Delta x_1} \tag{40}$$

$$\frac{\partial u_1}{\partial x_2}\bigg|_n^k \approx \frac{u_{1,(m,n+1)}^k - u_{1,(m,n)}^k}{\Delta x_2} \tag{41}$$

$$\frac{\partial u_1}{\partial x_2}\bigg|_s^k \approx \frac{u_{1,(m,n)}^k - u_{1,(m,n-1)}^k}{\Delta x_2} \tag{42}$$

The only term on the right hand side of the u_1 corrector equation is the pressure flux term. Because a staggered grid is used in this problem, the pressure nodes exist at the center of the East and West cell wall faces. A constant pressure is assumed at the cell walls. Accordingly, the pressure flux is approximated and discretized as shown in Eqs. 43 - 45.

$$\int_{\sigma} p n_1 d\sigma = \int_{\sigma_e} p d\sigma - \int_{\sigma_w} p d\sigma \tag{43}$$

$$\int_{\sigma_e} p d\sigma \equiv F^k_{p,e} \approx p_e \Delta x_2 = p_{(m+1,n)} \Delta x_2 \tag{44}$$

$$-\int_{\sigma_w} p d\sigma \equiv F^k_{p,w} \approx -p_w \Delta x_2 = -p_{(m,n)} \Delta x_2 \tag{45}$$

3.3.2 u_2 Momentum Equation

3.4 Solving the Pressure Poisson Equation

4 Temporal Discretization

5 Solution Procedure

6 Results

References

1. Ferziger, J. H. & Perić, M. *Computational Methods for Fluid Dynamics* 3rd ed. (Springer, Heidelberg, 2002).